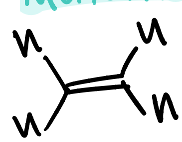
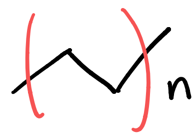
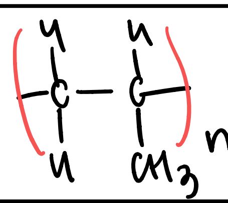
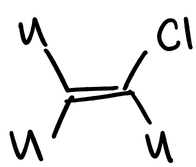
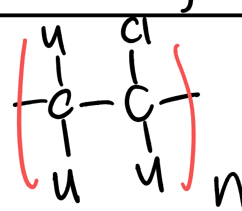
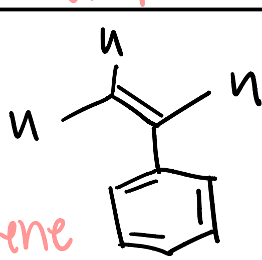
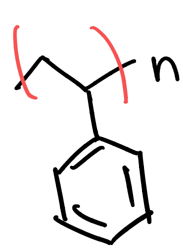
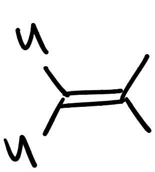
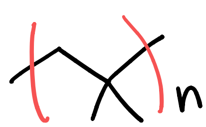
CHE 330 HW 1 Questions
Assigned 8/27/2025, Due 9/02/2025 11:59 PM

Polymers Basics

1. Read Chapter 1 of the course textbook ([Paul C. Hiemenz, Timothy P. Lodge - Polymer Chemistry-CRC Press (2007).pdf]) and the reading/viewing assignments that were posted on the UIC blackboard.

2. List five polymers that you can find in your daily life.
   a. For each polymer, please provide the name of the polymer, the chemical structure of the monomer, and the chemical structure of the polymer using the right notation.
   b. Explain why it is used for that application.

	Name	Monomer	Polymer	Application
1	polyethylene	 Ethylene		Used to make plastic bottles & bags. used b/c its light weight, cheap & flexible
2	polypropylene	 propylene		food containers b/c H is more rigid & heat resistant
3	poly vinyl chloride	 vinyl chloride		pipes b/c its strong, durable, & resistant to corrosion & chemicals long lasting in the air
4	polystyrene	 Styrene		Styrofoam b/c its rigid & can be formed into lightweight insulating material
5	polyisobutane			Used in tires & gum bases b/c its highly flexible and elastic

3. Explain the difference between monomer, dimer, oligomer, polymer.

- Monomer: smallest basic molecular unit that can join w/ others to form larger structures
  one
- Dimer: molecule made of 2 monomers chemically bonded
  two
- Oligomer: short repeating chain of monomers ~2-20 chains
  few
- Polymer: very long chain or network of repeating monomer units
  many

Polymer Classification

4. Would you consider a polymer of MW=25,000 a low polymer or a high polymer? Why?

This molecular weight is way above that of an oligomer / low polymer range making it a high polymer.
This MW suggests that it is a high polymer b/c a high polymer has a very large number of repeating units.

5. Identify five polymers that can be found in nature

- Cellulose: linear chain of glucose units
- Starch: branched glucose chains
- Proteins: polypeptide chains
- DNA
- Natural Rubber

6. Natural polymers are made by living organisms, while synthetic polymers are produced in the lab by chemical reactions. Explain a semi-synthetic polymer.

A semi-synthetic polymer is a polymer that is derived from a natural polymer but has been chemically modified to improve its properties.

7. List the 2 types of thermal responses of a polymer that we discussed in class. Describe them in your own words and list an example from class for each.

Thermoplastics: soften or melt when heated, harden when cooled. This allows them to be reshaped.

ex: water bottles

Thermo setting: Hardening is irreversible after they are heated & cannot be remelted.

ex: Rubber tires

8. Which has a slower conversion, chain growth or step growth formation? (Hint: Use the graph)

Step growth is slower. Chain growth reached the high monomer conversion much quicker.

9. Briefly discuss 2 differences between a copolymer and a homopolymer.

Properties

- Homopolymer's depends on 1 repeating unit
- copolymer can be tailored / combined making them versatile

Composition

- Homopolymer: single monomer species
- copolymer: At least 2 diff monomer species

10. Why do cellulose and starch, both made from glucose, have different properties? Describe how the differences in their structure at different levels (basic building blocks, chain arrangement, and overall 3D shape) affect properties like how well they dissolve in water, their crystallinity, and their rigidity.

1) Building blocks:

- starch uses same orientation linkages not as many H bonds, more water uptake, lower crystallinity, lower rigidity
- cellulose alternates linkages which forms strong H-bonds from aligned -OH groups, poor water dissolution, high crystallinity, high rigidity

Chain arrangement: starch is helical & highly branched which means it packs poorly. Easier water penetration.

Cellulose is linear & unbranched, it packs well & excludes water leading to higher crystallinity overall 3D shape:

- cellulose: extended ribbon like linear chains, very rigid
- starch: helical coils + branched clusters, softer granulated form